RURAL HEALTH · LABOR · ACCESS

Rural Worker Healthcare Friction Framework

Work Conditions → Five Healthcare Frictions → Delayed Care

Original five-friction structure · Development draft

Work conditions create friction before they create delayed care

WORK CONDITIONS

Agricultural calendar
Long working hours
Time-sensitive tasks
Household labor structure



1 **Labor Friction**
Long hours · cannot stop work

2 **Replacement Friction**
No reliable person can take over

3 **Time & Income Friction**
Half-day visit · lost farm or wage income

4 **Transportation Friction**
Distance · transit time · transfers

5 **Digital Friction**
Registration · department choice · dependence



DELAYED CARE

Postpone appointments
Continue working while unwell
Seek care only after function declines

The framework explains why healthcare may exist but still remain difficult to use.

Each friction has a distinct evidence signal and mechanism

The matrix separates what the data show from how interviews explain it.

Friction	Quantitative signal	Mechanism	Evidence strength	Framework use
Labor	75.4% often/sometimes lack time; >6h workers show more delay	Agricultural tasks require continuous, time-sensitive blocks	Strong	Explains time poverty
Replacement	No-replacement group: 85.7% delayed vs 74.5% with some replacement	A helper must know the work and assume responsibility	Strong	Explains inability to leave
Time & Income	74.1% hesitate when care takes half a day; income pressure common	Cost means missed harvest, orders, wages and family production	Mixed	Explains opportunity cost
Transportation	27.2% report transport barriers; audited transit can take 30–90 min	Distance becomes friction through transfers and uncertain return time	Moderate	Explains physical access
Digital	55.8% have weak registration ability; 60+ group is substantially higher	Problems occur at department choice, verification, insurance and results	Very strong	Explains navigation dependence

Evidence strength guides design priorities; it does not turn the framework into a validated clinical scale.

Journey 1 · Labor and replacement barriers compound each other

Grape grower in the harvest period · long workday · recurring lower-back pain

TRIGGER

Pain begins during a critical work window

Harvest cannot simply move to another day; tasks accumulate immediately.

FRICTION CHAIN

No reliable replacement for half a day

A family member may help briefly, but cannot fully take over the work.

OUTCOME

Keeps working until function declines

Care is delayed until pain affects sleep, walking or lifting.

BEST-FIT RESPONSE **Pre-busy-season clinic + replacement plan + musculoskeletal red flags**

Journey 2 · Digital and transport barriers turn one visit into a chain

Older agricultural worker · knee pain · depends on family for hospital navigation

TRIGGER

Needs care beyond the village station

The next step requires choosing a hospital, department and appointment.

FRICTION CHAIN

Family absent, registration unclear, transit long

Digital steps and a 30–90 minute transit journey reinforce each other.

OUTCOME

Postpones or settles for incomplete help

The formal service exists, but the pathway is not independently usable.

BEST-FIT RESPONSE **Fixed navigation contact + offline registration + route card**

Journey 3 · Opportunity cost can outweigh access

Middle-aged farm worker paid by output · can use a smartphone · transport available

TRIGGER

A manageable symptom appears on a paid workday

The worker can register and travel, so conventional access measures look favorable.

FRICTION CHAIN

Stopping work creates immediate income loss

Replacement does not restore the worker's wages or agricultural output.

OUTCOME

Care is delayed despite digital ability

The binding friction is the economic cost of a half-day visit.

BEST-FIT RESPONSE **Evening care + short visits + fewer repeat trips**

The framework converts diagnosis into intervention design

Different frictions require different solutions; a single health lecture cannot address all five.

Friction	Diagnostic question	Primary response	Operational design	Evaluation metric
Labor	Can the worker leave for 2–4 hours?	Schedule-compatible care	Village, evening or pre-busy services	Time away from work
Replacement	Who can reliably take over?	Replacement planning	Family/neighbor plan before appointment	Visits completed with coverage
Time & Income	What is lost during a half-day visit?	Reduce opportunity cost	Short appointments; one-stop checks	Work hours and trips saved
Transportation	How long and how many transfers?	Bring care closer or simplify travel	Route card, referral, transport support	Door-to-door travel time
Digital	Which step fails without help?	Persistent navigation support	Offline registration; fixed helper	Independent/assisted completion

The goal is not more information. It is less friction between health need and usable care.